

Ayushi Sehgal

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SUMMARY

Results-driven Computer Science student with **8.6 CGPA** and strong foundations in Data Structures, Algorithms, and Object-Oriented Programming. Experienced in **3+ full-stack and machine learning projects**, using Java, Spring Boot, React, and Python. Built scalable applications handling **100+ users/data instances** and developed ML models with **90%+ accuracy**. Seeking software engineering or AI/ML internship opportunities to apply technical and problem-solving skills.

TECHNICAL SKILLS

Programming Languages: Java, C++, Python, HTML, JavaScript(**500+ DSA problems solved**)

Web and Backend: Spring Boot, Node.js, Express.js(**3+ projects**), , REST APIs, Microservices

Frontend: React.js, HTML, CSS(**2+ responsive applications**)

Databases: MySQL, PostgreSQL, MongoDB(**3+ integrated projects**)

Tools and Platforms: Git, Docker, AWS, Maven

Machine Learning: Scikit-learn, TensorFlow, Pandas, NumPy (**2+ ML models built**)

Core Concepts: Data Structures & Algorithms, DBMS, Operating Systems, OOP, Agile/Scrum

Soft skills: Team Collaboration, Problem resolution, Effective communication, Flexible, Logical reasoning, Leadership initiative

PROJECTS

College News Portal

Sept 2024 – Jan 2025

Web Development Project

HTML,CSS,JavaScript,Express.js,Node.js

- Developed a **full-stack web application** using Node.js and Express.js to manage and publish news for **100+ college clubs and 500+ users**
- Designed **role-based access control (RBAC)** system enabling secure admin operations and reducing unauthorized access by **30%**
- Implemented **RESTful APIs** and dynamic content rendering, improving content accessibility by **40%**

Financial Dashboard

Feb 2025 – May 2025

Web Development Project

Node.js, Express.js, HTML,CSS,JavaScript,

- Built a **financial dashboard web application** using MVC architecture, improving code maintainability by **35%**
- Implemented **secure authentication and role-based access** for **3+ user roles**, ensuring **100% data privacy compliance**
- Structured backend logic to handle **real-time data updates**, improving response efficiency by **25%**

Cancer Detection using Machine Learning

July 2025 – current

Machine learning Project

Python

- Developed a **supervised machine learning classification model** using Scikit-learn achieving **92–96% accuracy** on the Wisconsin Breast Cancer dataset .
- Performed **data preprocessing and feature engineering** on **500+ data samples**, reducing dimensionality by **60% using PCA** .
- Compared **3+ algorithms (Logistic Regression, SVM, etc.)**, improving model performance by **15% through hyperparameter tuning**
- Applied **Explainable AI (SHAP, LIME)** to improve model interpretability by **40%**

RESEARCH SECTION

Big Data Management Framework

Sept 2024 – Jan 2025

- Conducted comparative analysis of **3 major big data frameworks (Hadoop, Spark, Flink)** improving system selection efficiency by **30%** .
- Designed and evaluated **2+ data processing models** for large-scale datasets (**10K+ records simulated**)
- Delivered **10+ technical presentations**, improving team understanding and decision-making efficiency

Study of Sentiment Analysis

July 2025 – November 2025

- Analyzed **1000+ social media data samples** using NLP techniques for sentiment classification
- Designed an end-to-end pipeline including **data collection, preprocessing, and classification**, improving processing efficiency by **35%**
- Identified **5+ key challenges** and proposed scalable solutions using big data tools

CERTIFICATION

- MathWorks **MATLAB Onramp** - Certified in core programming, data manipulation, and plotting techniques.
- Successfully completed the **NPTEL "Machine Learning"** certification course(12 week course)
- **AlgoUniversity** - Completed a rigorous university-level curriculum on **Graph Theory** under the mentorship of a Codeforces Master (Manas Kumar Verma), conquering **17 + complex problems involving advanced concepts** like shortest paths, network flows, and graph traversals.
- **HackerRank Certified in Python** (Basic), demonstrating proficiency in 100+ fundamental syntax, data structures, problem-solving and solving coding challenges on strings, collections, and algorithms.
- Completed **Udemy certification: "Java Full Stack Developer" by Chand Sheikh(50+ hours)**, covering Java, Spring Boot, React, and MySQL.
- Achieved **Oracle Cloud Infrastructure 2025 Certified Foundations Associate**, validating core cloud infrastructure and platform services expertise.
- Completed **Google's 'The Bits and Bytes of Computer Networking' certificate(40+ hours)** gaining in-depth knowledge of network protocols, infrastructure, and cloud-based networking solutions.

Virtual Job Simulations(Forage)

- Completed **JPMorgan Chase Software Engineering Virtual Experience (Forage)** by developing a **Spring Boot microservice integrating Apache Kafka, Spring Data JPA, and REST APIs** to process, validate, and persist high-volume financial transactions with end-to-end testing.
- Completed **AWS Solutions Architecture Virtual Experience (Forage)** by designing a **scalable, cost-efficient cloud architecture using AWS Elastic Beanstalk**, improving application performance and clearly communicating technical design and pricing to non-technical stakeholders.

WORK EXPERIENCE

SmartWeb Solutions(New Delhi)

January 2025 – April 2025

- Developed and maintained **10+ responsive web pages**, improving user experience and accessibility
- Collaborated in an **Agile team of 5+ members**, contributing to **2+ project sprints**
- Identified and resolved **20+ front-end bugs**, improving website performance by **30%** .
- Improved cross-browser compatibility across **5+ browsers/devices** .

EDUCATION

St.Joseph's Academy

Dehradun, Uttarakhand

- Scored a **93% in 10th grade** , showcasing strong academic achievement and building a solid groundwork for higher education.
- Attained a **90% in 12th grade**, highlighting sustained commitment to academics and strong grasp of core subjects like Physics, Chemistry, Maths and computer science.
- Not only in studies but I have also achieved excellence in dance, basketball, debating

Vellore Institute of Technology

Bhopal,Madhya Pradesh

- Currently pursuing a Bachelor of Technology (B. Tech) in Computer Science and Engineering, in the 6th semester with a commendable CGPA of **8.6/10**, reflecting strong academic achievement.
- Built a robust foundation in core programming languages including **Java and c++**, essential for **effective software development**.
- Developed proficiency in **data structures and algorithms**, significantly strengthening analytical and problem-solving capabilities.
- Gained comprehensive understanding of **operating systems** and **database management systems**, vital for efficient system architecture and data management.
- As the **club coordinator of the Dance Club** which is the most renowned club of our college, I have gained so much experience in organizing small as well as large scale events(**1000+ students**), I believe that having management skills is one of my strongest qualities..